

KHALED MOSHARRAF MUKUT, PH.D.



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🌐 kmmukut.pages.dev

About Me

Postdoctoral Research Fellow in Mechanical Engineering specializing in CFD, multiscale modeling, and high-performance computing. Proven track record in leading research on soot formation and combustion, with multiple publications and a distinguished research fellowship. Passionate about mentoring students and collaborating across disciplines to drive innovative solutions.

Technical Skills

Programming Languages: Python, C/C++, MATLAB, FORTRAN, HTML, Bash
Simulation Tools: OpenFOAM, LAMMPS, ANSYS, COMSOL, CONVERGE CFD
Engineering Tools: VS Code, PyCharm, SolidWorks, Tecplot, Paraview, Tableau
Technologies/Frameworks: Linux (HPC), Git, Docker, MPI/OpenMP

Professional Experience

Marquette University

Oct 2025 – Present

Postdoctoral Research Fellow

Milwaukee, WI

- Leading advanced research on microclimate modeling at a city-block scale using high-fidelity CFD.
- Developing frameworks for optimizing load balancing in large-scale Lagrangian-Eulerian simulations on HPC systems.

Marquette University

Aug 2017 – Aug 2025

Graduate Research Assistant

Milwaukee, WI

- Conducted research leading to 7 peer-reviewed journal publications and presented findings at 8 major conferences.
- Exhibited strong proficiency in multidisciplinary computational skills, encompassing CFD and molecular modeling.
- Instructed and graded undergraduate courses including Heat Transfer, Fluid Mechanics, and Thermodynamics.
- Mentored graduate students in essential research tools including Git, Linux, Python, and L^AT_EX.

Bangladesh University of Engineering & Technology

Mar 2016 – Aug 2017

Graduate Assistant

Dhaka, BD

- Mentored two undergraduate senior groups on molecular dynamics simulations and ANSYS modeling.
- Published 3 journal articles and presented at 1 international conference.

Education

Ph.D in Mechanical Engineering (Marquette University)

Aug 2025

Dissertation: Molecular simulations of soot formation and morphology.

Milwaukee, WI

MS in Mechanical Engineering (Marquette University)

May 2019

Thesis: Radiation effects on pollutant formation in spray combustion.

Milwaukee, WI

Honors and Awards

Richard W. Jobling Distinguished Research Fellowship | Marquette University

Feb 2023

Outstanding Research Assistant Award | Marquette University

Apr 2021

Dean's List Scholarship | BUET

2011 & 2012

Leadership & Service

Python Summer School | Bootcamp for high school seniors

Jul 2023, Jul 2025

- Guided 15 students to create their own air quality data visualizations using Python.

Entangled Air Exhibition | Art & Science Collaboration

Jan 2023 – May 2023

- Developed artwork and led outreach initiatives raising awareness about air quality.

President | Bangladeshi Student Association (BSAMU)

Jun 2021 – May 2023

- Led executive board of 10 members; organized cultural events and community service.

Selected Publications

- K. M. Mukut et al.: *Fuel* 406, 136692 (2026)
- T. P. Day, K. M. Mukut et al.: *Proc. Combust. Inst.* 41, 105821 (2025)
- A. Ganguly, K. M. Mukut et al.: *Aerosol Research* 3, 185–203 (2025)
- K. M. Mukut et al.: *J. Phys. Chem. A* 128, 5175 (2024)

- Full list: kmmukut.pages.dev/Publications.html

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